Hypothesis Testing

Control Group ⟷ Treatment Group

Before Manipulation ⟷ After Manipulation

Question: Has something changed

Website averages daily visits 10 372 μ
after marketing we sample 30 days n
and finds a sample mean 10 628 $\bar{X}$ avg. daily users
w/ sample SD = 402

Ⓐ Null Hypothesis (H_0): The mean has not changed & increase is due statistical fluctuations.

Ⓑ Alternative Hypothesis (H_A):
The mean has increased
too large of change for "random" → Null Hypo is False

Procedure

Ⓘ We assume Null Hypothesis H_0 is true
(average is still $\mu = 10\,372$)

Ⓘ Ⓘ We define a test statistic

$$Z = \frac{\text{Obs } \bar{X} - \text{expected avg.}}{\text{standard error}} = \frac{\bar{X} - \mu}{\hat{\sigma}/\sqrt{n}}$$

$$\frac{10\,628 - 10\,372}{402/\sqrt{30}} \approx 3.49 \text{ S.E.'s}$$

Ⓘ Ⓘ Ⓘ Compute the observed significance level p
we reject Null Hypothesis H_0

$P(Z \leq 3{,}49) = 0.998$

$p = 0.002$ p-value